

Irrationality of Cubic Threefolds after One Stabilization

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Abstract

We prove that $X \times \mathbf{P}^1$ is irrational for every smooth complex cubic threefold X . On the generic even quantum D -module, our invariant counts rank-two formal connection blocks whose centered leading Euler operator is nonzero square-zero and whose two formal exponent classes differ modulo the integers. Blowup and projective-bundle decompositions make this count birationally invariant because all low-dimensional centers contribute zero. It distinguishes $X \times \mathbf{P}^1$ from $\mathbf{P}^4$. The same count gives an irrationality criterion for smooth projective threefolds, the one-stabilization theorem for prime Fano threefolds of genus eight, and separation results on known universally CH_0 -trivial loci.

Keywords. cubic threefold; universal CH_0 -triviality; irrationality; quantum D -modules; formal exponents; block markers; weak factorization; Fano threefolds.

1 Introduction

Let $X \subset \mathbf{P}^4$ be a smooth complex cubic threefold. Clemens and Griffiths proved that X is irrational [6], but their intermediate-Jacobian obstruction does not directly survive on $X \times \mathbf{P}^1$: in a factorization of fourfolds, a surface center can contribute precisely the weight-three Hodge structure carried by X . We prove that the stabilized fourfold is nevertheless irrational.

Theorem 1.1 (One-step irrationality). *For every smooth complex cubic threefold X , the fourfold $X \times \mathbf{P}^1$ is irrational.*

The proof uses the generic even quantum D -module (QDM). Over an algebraic closure of its generic coefficient field, Euler multiplication splits the formal connection into blocks. We place those blocks in an effective commutative ledger. Iritani's blowup theorem and Iritani–Koto's projective-bundle theorem give additive operation formulas, with each copy of a blowup center retained as a separate occurrence. Weak factorization then makes every additive block marker that vanishes on the possible centers a birational invariant.

The marker used here detects a rank-two block whose centered leading Euler operator is nonzero square-zero and whose two formal exponent classes are distinct modulo the integers. Pairing horizontality makes the required elementary modification regular; flatness moves its residue by conjugation. For a cubic threefold, Beauville's quantum products give exponent representatives $-1/6$ and $-5/6$, with difference $2/3$. The marker vanishes on points, curves, and surfaces, doubles on product with $\mathbf{P}^1$, and vanishes on projective space. This proves the theorem.

The comparison inputs are Iritani's blowup decomposition and Iritani–Koto's projective-bundle decomposition [8, 9]. What is new here is their use as an occurrence-indexed additive ledger through weak factorization, together with the formal-exponent fold and its low-dimensional nullity. Cai computes the cubic quantum differential equation in a compatible

normalization [5]; that calculation neither constructs this birational marker nor proves the one-stabilization conclusion.

The same construction gives a threefold criterion without stabilization.

Corollary 1.2 (Threefold marker criterion). *Let Y be a smooth projective complex threefold. If its generic even QDM contains a block marked by (2.12), then Y is irrational.*

Indeed, the only centers in a weak factorization of threefolds have dimension at most one, where the marker vanishes, while the generic quantum product of $\mathbf{P}^3$ is semisimple. This is a QDM-side irrationality criterion. It is not identified here with semiorthogonal components or with a derived categorical rationality criterion.

The result also propagates through stabilized birational correspondences. Kuznetsov’s construction applies to an arbitrary smooth prime Fano threefold V of genus eight: it associates a smooth Pfaffian cubic threefold X and rank-two projective bundles over V and X related by a flop [10, Theorems 2.17–2.18]. This gives the following consequence.

Corollary 1.3 (One stabilization of V_{14}). *For every smooth prime Fano threefold V of genus eight, $V \times \mathbf{P}^1$ is irrational.*

The theorem also separates irrationality from universal CH_0 -triviality on several known loci of cubic threefolds. Universal CH_0 -triviality passes to a projective bundle by the projective-bundle formula for Chow groups.

Corollary 1.4 (Separation in codimension three). *There is a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds such that, for every member X , both X and $X \times \mathbf{P}^1$ are universally CH_0 -trivial, while $X \times \mathbf{P}^1$ is irrational.*

Corollary 1.5 (Fermat). *The fourfold*

$$\{x_1^3 + x_2^3 + x_3^3 + x_4^3 + x_5^3 = 0\} \times \mathbf{P}^1$$

is universally CH_0 -trivial and irrational.

Corollary 1.6 (Coprime-degree cubics). *Let X be one of the cubic threefolds of [13, Theorem 3.3]. Then $X \times \mathbf{P}^1$ admits unirational parametrizations of coprime degrees, is universally CH_0 -trivial, and is irrational.*

Voisin supplies the codimension-three locus [12, Theorem 4.5]; Colliot-Thélène’s separated-variable criterion applies to the Fermat cubic [7]; and Yang–Yu–Zhu construct the coprime-degree family [13, Theorems 1.1 and 3.3]. The last corollary settles their question at the first nontrivial stabilization, not at every stabilization.

Section 2 constructs the block ledger, proves its QDM operation formulas, defines the formal-exponent marker, computes the cubic block, and proves center nullity. Section 3 gives the threefold, genus-eight, and universal- CH_0 consequences.

2 QDM markers and one stabilization

We prove the theorem by applying the quantum D -module decomposition formulas directly. The common argument is stated first for an arbitrary additive marker. The rest of the section constructs the smallest marker needed for a cubic threefold and verifies its center nullity.

Here is the concrete invariant before the general construction. Split the generic even quantum D -module into formal connection blocks. A block contributes one when it has

rank two, its centered leading Euler operator is nonzero square-zero, and the two exponent classes obtained after the canonical elementary modification are distinct modulo $\mathbf{Z}$; every other block contributes zero. Their sum is I_{at} . The rest of the section shows that this count is intrinsic and additive, vanishes on all possible centers, equals one on a cubic threefold, doubles after product with $\mathbf{P}^1$, and vanishes on $\mathbf{P}^4$.

2.1 The one-blowup model

Let Y be smooth and projective. Pass to numerical Novikov variables, set the odd bulk variables to zero, and restrict the carrier of the quantum connection to $H^{\text{ev}}(Y)$. The last restriction is essential: even bulk alone still leaves the odd cohomology in the fibre. Let K_Y be the generic coefficient field of this even QDM and fix an algebraic closure $\bar{K}_Y$. All spectral blocks below are formed only after base change to $\bar{K}_Y$. Thus later scalar extensions do not split a primary factor that was previously counted as one block.

Let U be multiplication by the Euler field on the even carrier. Its generalized eigenspaces over $\bar{K}_Y$ are summands of the full formal connection, rather than merely summands of U :

Proposition 2.1 (Formal spectral connection splitting). *Suppose the leading operator of a finite free formal QDM has block-diagonal form $U_i = \lambda_i I + N_i$, where each N_i is nilpotent and $\lambda_i - \lambda_j$ is a unit for $i \neq j$. There is a unique gauge $G = I + O(z)$, regular in z and block off-diagonal in positive orders, which makes the z -connection block diagonal. In this gauge every bulk and Novikov direction is block diagonal as well. The horizontal pairing is block diagonal and nondegenerate on each factor.*

Proof. At order z^{n-1} , the (i, j) -entry of the gauge equation is a Sylvester equation

$$U_i X - X U_j = B_{n,ij}.$$

Its operator is $(\lambda_i - \lambda_j)I$ plus the nilpotent operator $X \mapsto N_i X - X N_j$, hence is invertible. Recursion gives the normalized gauge, and the same equation with zero right-hand side proves uniqueness.

Choose frames of the generalized-eigenspace subbundles after the splitting extension. If an off-diagonal block of a base-direction connection had a first nonzero Laurent coefficient Xz^m , the lowest off-diagonal coefficient of flatness would be $U_i X - X U_j = 0$, contrary to Sylvester invertibility. Thus every base direction preserves the same factors. Applied coefficientwise to pairing horizontality, the identical argument kills the off-diagonal pairing blocks. Nondegeneracy of the total pairing then makes every diagonal restriction nondegenerate. $\square$

We call these factors the *generic even QDM blocks* of Y . A basic block retains its connection, pairing, and centered Euler endomorphism, but not an absolute cohomological grading. A block type may require additional marked data. If it retains grading, morphisms may carry a recorded integral suspension and its marker fold must be invariant under that suspension. The atomic type below forgets absolute grading. Its morphisms preserve all retained data and are regular connection isomorphisms after scalar extension and formal change of bulk coordinates. In particular, their gauges use neither negative nor fractional powers of the connection variable z .

Fix a block type $\mathcal{T}$. Let $\Pi_{\mathcal{T}}$ be the regular-isomorphism groupoid of blocks of that type and put

$$\mathcal{L}_{\mathcal{T}} := \pi_0 \text{Sym}^{\sqcup}(\Pi_{\mathcal{T}}),$$

the effective free commutative monoid on their isomorphism classes. Direct sum gives each Y a block ledger $\mathcal{B}_{\mathcal{T}}(Y) \in \mathcal{L}_{\mathcal{T}}$. A *marker fold* is a commutative-monoid homomorphism

$$w : \mathcal{L}_{\mathcal{T}} \longrightarrow A, \quad I_w(Y) := w(\mathcal{B}_{\mathcal{T}}(Y)),$$

where A is an effective commutative monoid. No group completion is made: it can destroy useful folds, including Boolean ones.

The two specializations use the following notation.

Symbol	Meaning
$\Pi_{\mathcal{T}}$	regular-isomorphism groupoid of typed QDM blocks
$\mathcal{L}_{\mathcal{T}}$	$\pi_0 \text{Sym}^{\sqcup}(\Pi_{\mathcal{T}})$, the effective block monoid
$\mathcal{B}_{\mathcal{T}}(Y)$	block ledger of Y
$w : \mathcal{L}_{\mathcal{T}} \rightarrow A$	additive marker fold into an effective monoid
ω_j	the j -th center occurrence in one blowup
$c_w(\omega_j)$	marker of that actual center occurrence

Table 1: Notation for the occurrence-indexed marker ledger.

Consider one blowup $\beta : \tilde{Y} = \text{Bl}_Z Y \rightarrow Y$ along a smooth center of codimension $c \geq 2$. Iritani's comparison has one ambient summand and $c - 1$ center summands. We retain the occurrence data of each center summand: for the j -th, record

$$\omega_j = (\beta, Z, j, \varsigma_j, \chi_j), \quad 0 \leq j \leq c - 2,$$

where ς_j is Iritani's center reconstruction parameter and χ_j is the induced numerical Novikov specialization. Its correction is $c_w(\omega_j) := w(\mathcal{B}_{\mathcal{T}}(Z; \omega_j))$. Equality is required only after applying w ; distinct block multisets may have the same marker.

Theorem 2.2 (Occurrence-indexed marker ledger). *Let $w : \mathcal{L}_{\mathcal{T}} \rightarrow A$ be a marker fold unchanged by the allowed regular scalar extensions and formal coordinate changes. Suppose*

$$I_w(\mathbf{P}_Y(V)) = rI_w(Y), \quad \text{rk } V = r, \quad (2.1)$$

$$I_w(\text{Bl}_Z Y) = I_w(Y) + \sum_{j=0}^{c-2} c_w(\omega_j), \quad \text{codim}_Y Z = c \geq 2. \quad (2.2)$$

If every actual smooth center occurrence of dimension at most $d - 2$ has zero correction, then I_w is a birational invariant of smooth projective d -folds.

Proof. Projective weak factorization writes a birational map of smooth projective d -folds as a zigzag of blowups and blowdowns along smooth centers of codimension at least two [1, Theorem 0.3.1]. By (2.2) and center nullity, the marker is unchanged at each step; the equalities therefore telescope. Disconnected centers are treated componentwise, while a divisor blowup is an isomorphism. $\square$

Equivalently, the marker descends on object sets:

$$\begin{array}{ccc} \text{Ob}(\text{SmProj}_d) & \xrightarrow{I_w} & A \\ \downarrow & & \parallel \\ \text{Ob}(\text{SmProj}_d)/\sim_{\text{bir}} & \xrightarrow{\bar{I}_w} & A. \end{array} \quad (2.3)$$

This square reformulates the proof; it is not an additional input.

2.2 The QDM operation providers

Four compatibility facts place the comparison theorems in the preceding framework. First, the intrinsic ample-adic completion and the Laurent neighborhood of a fibre or exceptional cusp need not map to one another. Iritani's display (1.1) supplies the required injective maps of graded completed coefficient rings [8, (1.1)]. For projective bundles the common source is $\mathbf{C}[z, q][[Q, \hat{\tau}]]$; on the blowup side one uses the corresponding graded source and, for center terms, the reduced image ring of Iritani's Remark 5.6. Embed the generic fields of these common images in a fixed algebraically closed overfield and form every primary block there. Base change after this splitting extension preserves the rank, nilpotent Jordan type, regular formal gauge, and formal exponent classes of each block; forming primary factors over a nonsplitting field before extension would not.

Second, the comparison is restricted to the even carrier, not merely to the even bulk base. The supermodule conventions and the pull-push and Fourier formulas in Iritani's Section 5.8 and Iritani–Koto's construction are parity-preserving. After the odd bulk variables are set to zero, the comparison and its inverse therefore restrict to mutually inverse maps on H^{ev} . This is the QDM on which the atomic ledger is formed.

Third, the comparisons are regular in z . Iritani's Theorem 5.18 and Iritani–Koto's Theorem 5.1 are respectively defined over rings of the form

$$\mathbf{C}[z]((q_{\text{exc}}^{-1/\mathfrak{s}}))[[Q, \hat{\tau}]], \quad \mathbf{C}[z]((q_{\text{fib}}^{-1/r'})[[Q, \hat{\tau}]],$$

where Iritani–Koto take $r' = r$ when $r - 1$ is even and $r' = 2r$ when $r - 1$ is odd. Here $\mathfrak{s} \in \{c - 1, 2(c - 1)\}$ is Iritani's parity-dependent ramification index from [8, (5.11)]. Their inverses use no negative powers of z ; see also [9, Remark 5.3]. The maps are homogeneous but need not have degree zero: the projective-bundle map has degree $-(r - 1)$, while Iritani's ambient and center components have degrees 0 and $-c$, respectively [9, Theorem 5.1(3)][8, Theorem 5.18(3)]. Because the atomic block type does not retain absolute grading, these shifts do not change its component. For a graded refinement they are recorded as integral suspensions, and the operation formula applies only to a fold invariant under those suspensions.

Fourth, Iritani's center map need not be injective [8, (5.15)]. After numerical reduction it has the form

$$Q_Z^d \longmapsto Q^{i_*d} q_{\text{exc}}^{-\rho_Z \cdot d/(c-1)}. \quad (2.4)$$

Here $\rho_Z = c_1(N_{Z/Y})$. Choose divisors $D_1, \dots, D_\rho$ separating $N_1(Z)$ and retain the characters $\exp(\sum_k (D_k \cdot d) s_k)$. Every fibre of (2.4) is finite: an ample degree from Y bounds the lattice points in the corresponding slice of $\overline{NE}(Z)$. On a fibre of size n , a generic one-parameter restriction gives distinct exponentials. Their derivatives of orders $0, \dots, n - 1$ form an invertible Vandermonde matrix, so the character-tagged map is injective. The resulting base change is the following.

Lemma 2.3 (Faithful center base change). *Let $R_{Z, \text{src}}^{\text{red}}$ be the coefficient domain whose $\mathbf{C}[z]$ -extension is Iritani's divisor-equation source ring $\mathbf{C}[z][[Q_Z e^{\sigma^{(2)}}, \sigma']][\sigma^0]$, and let $R_{\text{cmp}, j}^{\text{red}}$ be its completed image coefficient domain in the j -th common comparison ring [8, Remark 5.6]. After adjoining independent divisor variables $s_1, \dots, s_\rho$, the continuous map $\iota_j : R_{Z, \text{src}}^{\text{red}}[[s]] \rightarrow R_{\text{cmp}, j}^{\text{red}}[[s]]$ defined by*

$$Q_Z^d \exp\left(\sum_k (D_k \cdot d) s_k\right) \longmapsto Q^{i_*d} q_{\text{exc}}^{-\rho_Z \cdot d/(c-1)} \exp\left(\sum_k (D_k \cdot d) s_k\right) \quad (2.5)$$

is injective. It induces a field embedding

$$\text{Frac}(R_{Z, \text{src}}^{\text{red}}[[s]]) \hookrightarrow \text{Frac}(R_{\text{cmp}, j}^{\text{red}}[[s]]).$$

After this field extension, each center summand in Iritani's comparison is the faithful scalar extension, followed by the formal coordinate change ς_j , of the intrinsic generic numerical even QDM of Z .

Proof. Filter both completed rings by an ample degree pulled back from Y . In each bounded quotient, every fibre of the untagged monomial map is finite. On such a fibre choose an integral linear combination of the D_k that separates its curve classes. Restriction to the corresponding one-parameter subgroup turns a putative kernel element into a finite sum of distinct exponentials; its first n derivatives give a Vandermonde matrix, so every coefficient is zero. The map is therefore injective on every quotient and, by separated completeness, on the inverse limit.

Remark 5.6 of [8] defines the center connection over exactly the image of this divisor-equation-reduced ring, rather than over the bare noninjective map (5.15). Base-change the comparison of [8, Theorem 5.18] and its inverse along (2.5); their two identities remain identities. The resulting extension of fraction fields is faithfully flat. Finally, the invertible combined bulk Jacobian in Theorem 5.18(7) identifies the retained divisor coordinates with the center coordinates and supplies the formal reparametrization ς_j . $\square$

Proposition 2.4 (QDM operation ledgers). *Let a marker fold be preserved by the regular scalar extensions and formal coordinate changes described above, and by every recorded grading suspension if its block type retains grading. The projective-bundle and blowup QDM comparisons on the even carriers then supply (2.1) and (2.2); each center marker is evaluated at its actual occurrence ω_j .*

Proof. For a rank- r bundle, Iritani–Koto's Theorem 5.1 gives r copies of the base QDM after Laurent extension [9, Theorem 5.1]. Its invertible joint bulk Jacobian gives independent unit coordinates on the r target factors. In the j -th factor, changing its unit coordinate adds an independent scalar $u_j I$ to Euler multiplication. For two finite spectra, the resultant of their translated characteristic polynomials is a nonzero polynomial in $u_i - u_j$, with leading coefficient one. Hence the spectra of distinct factors are disjoint at the generic point: their primary blocks do not fuse when the direct sum is formed. Regularity, even-carrier restriction, the fixed splitting field, and suspension invariance therefore carry the marker blockwise across the comparison, and additivity gives (2.1). Tensoring the bundle by a line bundle, when needed for the global-generation convention, changes neither the projective bundle nor the intrinsic Novikov embedding [9, Remark 1.2 and Remark 5.2].

For a codimension- c blowup, Iritani's Theorem 5.18 gives one ambient QDM and $c - 1$ center QDMs [8, Theorem 5.18]. Parts (4), (6), and (7) give the comparison asymptotics, the leading reconstruction parameters, and the invertible combined bulk Jacobian, respectively. The independent unit coordinates supplied by part (7) give the same nonzero resultant argument, so the ambient and occurrence-indexed center spectra are pairwise disjoint generically. Regularity, even-carrier restriction, the fixed splitting field, suspension invariance, and Lemma 2.3 for each center copy then give (2.2). Each comparison is evaluated over its own common coefficient spine; the proof never composes Laurent comparisons recursively. $\square$

The QDM comparison need not identify the two ledger elements before applying the fold. The relevant block-ledger operation is instead the additivity square

$$\begin{array}{ccc} \mathcal{L}_{\mathcal{T}} \times \mathcal{L}_{\mathcal{T}} & \xrightarrow{+} & \mathcal{L}_{\mathcal{T}} \\ w \times w \downarrow & & \downarrow w \\ A \times A & \xrightarrow{+} & A. \end{array} \quad (2.6)$$

At marker level, Proposition 2.4 supplies the resulting equalities in A ; its blowup statement also retains the occurrence metadata used by Theorem 2.2.

2.3 The rank-two formal-exponent marker

Let $\mathcal{T}_{\text{at}}$ be the block type on the H^{ev} -carrier retaining the connection, pairing, and centered Euler endomorphism, with no absolute grading, marked bulk section, or loop framing, and specialize to $A = \mathbf{N}$. Let E be a generic even block with centered connection

$$z\partial_z y = A(z)y, \quad A(z) = z^{-1}N + A_0 + zA_1 + z^2A_2 + \cdots. \quad (2.7)$$

Assume $\text{rk } E = 2$, $N \neq 0$, and $N^2 = 0$. Then $L = \text{im } N = \ker N$ is a line. Define

$$E^\sharp = \{s \in E : s \bmod z \in L\}. \quad (2.8)$$

Lemma 2.5 (The regular coefficient preserves the nilpotent line). *Horizontality of the Poincaré pairing implies $A_0L \subset L$.*

Proof. Separated spectral factors are orthogonal for the horizontal Poincaré pairing: coefficientwise horizontality kills every off-diagonal block by an invertible Sylvester operator. Hence the restriction to E is nondegenerate. Write it as $P(z) = P_0 + zP_1 + \cdots$. Horizontality gives

$$z\partial_z P(z) + A(z)^T P(z) + P(z)A(-z) = 0.$$

The z^{-1} coefficient gives $N^T P_0 = P_0 N$. Thus L is isotropic and, in dimension two, $L^\perp = L$. The constant coefficient is

$$A_0^T P_0 + P_0 A_0 + N^T P_1 - P_1 N = 0.$$

Evaluating this identity on (x, x) , where $x \in L = \ker N$, gives $(A_0 x, x) = 0$. Hence $A_0 x \in L^\perp = L$. $\square$

Choose a frame with $L = \langle e_1 \rangle$ and $Ne_2 = \nu e_1$, $\nu \neq 0$. The modified lattice has frame e_1, ze_2 , so

$$A^\sharp = S^{-1}AS - S^{-1}z\partial_z S, \quad S = \text{diag}(1, z). \quad (2.9)$$

The term $z^{-1}N$ becomes regular. The only possible new pole is the E_{21} -coefficient of A_0 , which vanishes by Lemma 2.5. Thus $A^\sharp = R + zR_1 + \cdots$. Define

$$\delta^\sharp(E) := (\text{tr } R)^2 - 4 \det R. \quad (2.10)$$

Proposition 2.6 (Rank-two rigidity). *Under formal change of even bulk coordinates, the modified connection is regular in every base direction and*

$$\partial R = [G_\partial, R] \quad (2.11)$$

for a regular matrix G_∂ . Consequently $\partial(\delta^\sharp) = 0$. Regular connection isomorphisms preserve the domain and value of $\delta^\sharp$.

Proof. Write a centered base equation as $\partial y = (z^{-1}C_\partial + C_{\partial,0} + O(z))y$. The z^{-2} coefficient of flatness gives $[N, C_\partial] = 0$, and the trace of the z^{-1} coefficient gives $\text{tr } C_\partial = 0$; here the base equation has been scalar-centered. The commutant of a nonzero rank-one nilpotent 2×2 matrix is the coefficient-field span of I and N , so $C_\partial = q_\partial N$. The same z^{-1} coefficient gives

$$\partial N = -q_\partial N + [N, q_\partial A_0 - C_{\partial,0}].$$

Writing $B = q_{\partial}A_0 - C_{\partial,0}$, one has $\partial(N^2) = -2q_{\partial}N^2 + [N^2, B]$; uniqueness for the formal coefficient recursion preserves $N^2 = 0$, while the nonzero leading value prevents N from becoming the zero formal section. Thus formal bulk transport preserves the nonzero square-zero orbit and its image line. After modification, the only possible base pole is $z^{-1}kE_{21}$. If $R = \begin{pmatrix} a & \nu \\ c & d \end{pmatrix}$, the z^{-1} flatness equation is $kE_{21} + [R, kE_{21}] = 0$. Its diagonal entries are νk and $-\nu k$, so $k = 0$. The constant term of the flatness equation is now (2.11); trace and determinant are constant under infinitesimal conjugation. A regular isomorphism carries N , L , and the modified lattice to their counterparts and conjugates the residue. $\square$

Proposition 2.7 (Exponent interpretation). *If ρ_1, ρ_2 are the eigenvalues of R , their classes modulo $\mathbf{Z}$ are the formal exponent classes of the original centered block and*

$$\delta^{\sharp} = (\rho_1 - \rho_2)^2.$$

Proof. The elementary modification uses integral powers of z , so it preserves formal exponent classes modulo $\mathbf{Z}$. The modified connection is regular singular with residue R ; the regular-singular classification identifies the exponent classes with the residue eigenvalues modulo $\mathbf{Z}$ [11, Chapter 3]. The displayed identity is the quadratic discriminant formula. $\square$

For such a block E , write ρ_1, ρ_2 for the eigenvalues of its modified residue in the fixed algebraic closure $\overline{K}$ of the generic coefficient field. Define the fold on each block by

$$w_{\text{at}}(E) = \begin{cases} 1, & \text{rk } E = 2, N \neq 0, N^2 = 0, [\rho_1] \neq [\rho_2] \text{ in } \overline{K}/\mathbf{Z}, \\ 0, & \text{otherwise,} \end{cases} \quad (2.12)$$

and extend it additively to $w_{\text{at}} : \mathcal{L}_{\mathcal{T}_{\text{at}}} \rightarrow \mathbf{N}$. Put $I_{\text{at}}(Y) = w_{\text{at}}(\mathcal{B}_{\mathcal{T}_{\text{at}}}(Y))$. Propositions 2.6 and 2.7 make this a lawful marker for Proposition 2.4: conjugation preserves the two residue eigenvalues, while changes of regular-singular lattice and recorded suspensions shift exponents only by integers. The condition $\delta^{\sharp} \neq 0$ is insufficient: it says only $\rho_1 \neq \rho_2$ and therefore also marks resonant blocks.

2.4 The cubic block

Let $X \subset \mathbf{P}^4$ be a smooth cubic threefold, let $P = H|_X$, and put $q = Q^{\ell}$ for the line class and $r = (3q)^{1/2}$. Beauville's formulas give [3, main theorem and (2.1)–(2.3)]

$$P \star P = P^2 + 6q, \quad P \star P^2 = P^3 + 15qP, \quad P \star P^3 = 6qP^2 + 36q^2. \quad (2.13)$$

Thus, in the basis $(1, P, P^2, P^3)$,

$$K_X = 2 \begin{pmatrix} 0 & 6q & 0 & 36q^2 \\ 1 & 0 & 15q & 0 \\ 0 & 1 & 0 & 6q \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad G_X = \frac{1}{2} \text{diag}(3, 1, -1, -3), \quad (2.14)$$

and $z^2 \partial_z S = (K_X + zG_X)S$. This agrees with [5, Section 3].

Proposition 2.8 (Small-even cubic block reduction). *The characteristic and minimal polynomials of K_X are both $T^2(T^2 - 108q)$. After adjoining r , its eigenvalues are $6r, -6r, 0, 0$, and the zero generalized eigenspace is one rank-two Jordan block. A normalized gauge regular in z gives*

$$\begin{aligned} z^2 \partial_z \tilde{S}_+ &= (6r + O(z^2)) \tilde{S}_+, \\ z^2 \partial_z \tilde{S}_- &= (-6r + O(z^2)) \tilde{S}_-, \\ z^2 \partial_z \begin{pmatrix} \tilde{S}_3 \\ \tilde{S}_4 \end{pmatrix} &= [J_0 + zD_0 + z^2E_0 + O(z^3)] \begin{pmatrix} \tilde{S}_3 \\ \tilde{S}_4 \end{pmatrix}, \end{aligned} \quad (2.15)$$

where

$$J_0 = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}, \quad D_0 = \begin{pmatrix} -19/18 & 0 \\ 0 & 19/18 \end{pmatrix}, \quad E_0 = \begin{pmatrix} 0 & -14/(81r^2) \\ -8/81 & 0 \end{pmatrix}. \quad (2.16)$$

Proof. The matrix

$$C = \begin{pmatrix} 6r^3 & -6r^3 & 0 & -7r^2 \\ 7r^2 & 7r^2 & -2r^2 & 0 \\ 3r & -3r & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}, \quad \det C = -486r^5,$$

puts K_X into

$$C^{-1}K_X C = \begin{pmatrix} 6r & 0 & 0 & 0 \\ 0 & -6r & 0 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Partition as $1 \mid 1 \mid 2$. At each positive order in z , the off-diagonal gauge coefficient is the unique solution of a Sylvester equation between blocks with disjoint scalar spectra. The gauge and inverse are regular in z ; explicitly, it has the normalized form

$$A(z) = I + \sum_{n \geq 1} A_n z^n, \quad A_n \text{ block-off-diagonal.} \quad (2.17)$$

Direct multiplication through order z^2 gives (2.16). $\square$

For the zero block, $N = J_0 = 2E_{12}$. With $S = \text{diag}(1, z)$,

$$R_X = 2E_{12} + D_0 - \frac{8}{81}E_{21} - \text{diag}(0, 1) = \begin{pmatrix} -19/18 & 2 \\ -8/81 & 1/18 \end{pmatrix}. \quad (2.18)$$

Its characteristic polynomial is $(T + 1/6)(T + 5/6)$. Thus the two exponent classes differ by $2/3 \notin \mathbf{Z}$, so the block is marked; as a numerical check,

$$\boxed{\delta^\#(X) = \frac{4}{9}}, \quad I_{\text{at}}(X) = 1. \quad (2.19)$$

2.5 Center nullity through dimension two

Proposition 2.9 (Low-dimensional nullity). *For every smooth projective Z of dimension at most two and every actual QDM comparison occurrence ω , $I_{\text{at}}(Z; \omega) = 0$.*

Proof. A point has one rank-one block. For $\mathbf{P}^1$, the relation $p \star p = Qe^{t_p}$ gives two rank-one generic blocks.

Let C have genus $g \geq 1$. There is no nonconstant genus-zero map to C , so its even big quantum product is classical. With $\int_C p = 1$ and $a = 2 - 2g$, its centered even connection is

$$z\partial_z Y = \left[z^{-1}aE_{21} + \text{diag}(1/2, -1/2) \right] Y.$$

For $g = 1$, $N = 0$. For $g > 1$, the modification $S = \text{diag}(z, 1)$ gives residue $aE_{21} - \frac{1}{2}I$, of discriminant zero. Its two eigenvalues, and therefore its two exponent classes, are equal. Thus every curve has zero marker.

Let S be a surface with K_S nef. For a quantum-product term with input degree b , non-scalar Euler degree $a \geq 2$, curve class d , and nonunit even bulk degrees $e_\nu \geq 2$, the dimension axiom gives output degree

$$a + b - 2c_1(S) \cdot d + \sum_{\nu} (e_\nu - 2) \geq b + 2.$$

The centered Euler operator strictly raises ordinary degree. Its only block is the whole even cohomology, of rank $b_0 + b_2 + b_4 \geq 3$, so its marker is zero.

A minimal surface with non-nef canonical class is $\mathbf{P}^2$ or a ruled surface $\mathbf{P}_C(V)$ [4, Chapter VI]. The relation $H^{*3} = Q$ makes $\mathbf{P}^2$ generically semisimple. Formula (2.1) and the curve case give zero for ruled surfaces. Every nonminimal surface is obtained from a minimal one by point blowups; formula (2.2) adds only zero point terms. This proves intrinsic nullity. Lemma 2.3 identifies every actual center term with a faithful scalar extension of the corresponding intrinsic generic block, and the marker is unchanged by that extension. $\square$

2.6 The projective endpoint and the contradiction

Formula (2.1) applied to $X \times \mathbf{P}^1 = \mathbf{P}_X(\mathcal{O}_X^{\oplus 2})$ gives

$$I_{\text{at}}(X \times \mathbf{P}^1) = 2. \tag{2.20}$$

On the other hand, over the generic numerical Novikov field $\Lambda = \text{Frac } \mathbf{C}[[Q]]$, $QH^\bullet(\mathbf{P}^4) = \Lambda[H]/(H^5 - Q)$. At $Q \neq 0$, multiplication by $5H$ has five distinct eigenvalues after algebraic closure. Thus

$$I_{\text{at}}(\mathbf{P}^4) = 0. \tag{2.21}$$

Proof of Theorem 1.1. Proposition 2.9 kills every actual center correction in a weak factorization of smooth projective fourfolds. Hence Theorem 2.2 makes I_{at} birationally invariant. If $X \times \mathbf{P}^1$ were rational, then

$$2 = I_{\text{at}}(X \times \mathbf{P}^1) = I_{\text{at}}(\mathbf{P}^4) = 0,$$

a contradiction. $\square$

3 Consequences

Proof of Corollary 1.2. In dimension three, every center in projective weak factorization has dimension at most one. The curve part of Proposition 2.9 therefore makes I_{at} birationally invariant. The generic quantum product of $\mathbf{P}^3$ is semisimple, so $I_{\text{at}}(\mathbf{P}^3) = 0$. A marked block gives $I_{\text{at}}(Y) > 0$, ruling out rationality. $\square$

Proof of Corollary 1.3. Kuznetsov's flop gives a birational map between rank-two projective bundles $\mathbf{P}_V(U)$ and $\mathbf{P}_X(E^*)$ over V and its associated Pfaffian cubic threefold X [10, Theorems 2.17–2.18]. Each projective bundle is birational to the product of its base with $\mathbf{P}^1$. Hence $V \times \mathbf{P}^1$ is birational to $X \times \mathbf{P}^1$, which is irrational by Theorem 1.1. $\square$

Proof of Corollary 1.4. Voisin produces a nonempty countable union of subvarieties of codimension at most three in the moduli space of smooth cubic threefolds along which the primitive minimal class $\Theta^4/4!$ is algebraic, hence along which CH_0 is universally trivial [12, Theorem 4.5 and Lemma 4.6]. The projective-bundle formula passes universal CH_0 -triviality to the product, and Theorem 1.1 supplies its irrationality. $\square$

Proof of Corollary 1.5. The Fermat equation is a sum of five forms in separated single variables, so [7] gives universal CH_0 -triviality; the projective-bundle formula passes it to the product. Irrationality follows from Theorem 1.1. $\square$

Proof of Corollary 1.6. The cubics of [13, Theorem 3.3] carry unirational parametrizations of degrees two and three, and [13, Corollary 3.5] keeps both degrees on $X \times \mathbf{P}^1$. Parametrizations of coprime degrees give a decomposition of the diagonal and hence universal CH_0 -triviality [13, Section 1]. Theorem 1.1 gives irrationality. $\square$

The marker operation formula has the same additive shape as Bittner’s blowup relation [2, Theorem 3.1]. More precisely, if an additive extension of a marker is defined on the Bittner group, then

$$I(\mathrm{Bl}_Z Y) - I(\mathbf{P}_Z(N_{Z/Y})) = I(Y) - I(Z)$$

follows from $I(\mathrm{Bl}_Z Y) = I(Y) + (c - 1)I(Z)$ and $I(\mathbf{P}_Z(N_{Z/Y})) = cI(Z)$. The effective $\mathbf{N}$ -valued fold used here is not itself such a group-valued motivic measure: no subtraction or multiplicativity has been constructed. The parallel explains why the center bound is decisive without claiming a Grothendieck-ring factorization that the proof does not supply.

All results above concern one product stabilization. No statement about a second or arbitrary stabilization is implicit.

AI assistance disclosure

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Numbered statements are cited from the exact versions listed here. Where both a journal and a preprint version appear, the numbering is that of the preprint.

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